

High Pressure Industrial Water (HPIW) B-4400

UESC LED Lighting Project

Stennis Space Center, MS

1.0 STATEMENT OF WORK

The scope of this project is to provide a Firm/Fixed pricing proposal for the energy conservation measures (ECM) to upgrade lighting at HPIW (B-4400). This Scope of work is based on the Preliminary Feasibility Audit (PFA) execute by MPC for this project and incorporates:

A. Design Documentation:

- I. Photometric Drawings – Normal and Emergency Lighting
- II. Lighting Schedule
- III. Electrical Site Plan
- IV. One Line Diagram
- V. Energy saving calculations
- VI. Safety Plan
- VII. Preliminary Environmental Survey
- VIII. Life Cycle Cost Analysis “LCCA” calculations
- IX. UFGS Construction Spec Package (if applicable)
- X. Commissioning Plan
- XI. Training

B. HPIW LED Lighting Upgrades Firm/Fixed Construction

2.0 CONSTRUCTION REQUIREMENTS

Comply with the most stringent requirements of NASA Standards, SSC Safety Procedures, Requirements, and Standards; the NFPA family of codes, especially National Electrical Code (NEC); American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE) guidelines; International Building Code (IBC); Institute of Electrical and Electronics Engineers (IEEE) standards; Environmental Protection Agency (EPA); Occupational Safety and Health Administration (OSHA).

3.0 SAFETY AND SECURITY

For the safety of staff, customers, and visitors, the Contractor shall be responsible for installing and maintaining temporary construction barriers. These barriers must be maintained throughout the life of the construction project. The Contractor will be given room for material laydown; however, NASA is not responsible for the safety or security of the materials or equipment. The project is being installed in an active building complex, and the Contractor shall take all reasonable steps to ensure a minimal amount of interference to those facilities. The Contractor and all personnel must report to the SSC Reception Center to obtain all necessary badging for access to the site.

4.0 CONTRACTOR PARKING

Contractors may park in any legal parking place. All Contractors must follow SSC requirements for driving and parking.

5.0 ASSUMPTION OF RISK

The Contractor shall assume all risk during construction; any losses from manmade hazards will be the responsibility of the Contractor. The Contractor is responsible for all risks for incidental damage to adjacent structures. The Contractor shall be responsible for all repairs and repair costs associated with damaged property at no cost to the Government. Assumption of risk shall also include risks associated with downtime for testing and/or other requests per Government discretion and schedule risks due to weather events. Furthermore, the Contractor is fully responsible for its own design and construction and is liable for its faults.

6.0 QUALITY CONTROL

NASA, the Contracting Officer (CO), or their designated agent shall have the right to inspect all materials and construction or installation methods to ensure they are per code, design, and industry standard. Design submittals, material cut sheets, and specifications shall be submitted as required at the design reviews. Closeout submittals shall include specifications, manufacturer's data, Operation and Maintenance (O&M) manuals, warranties, calculations, as-built drawings showing all utilities, Photometric files, and final drawings.

6.1 DESIGNER OF RECORD

The DB Contractor shall have within its organization a Designer of Record for each area of design deliverable that requires approval. Designer of Record shall be a licensed Professional Electrical Engineer registered in the State of Mississippi or an approved equivalent.

DESIGN DELIVERABLE FORMATS

- Drawing electronic files shall be in AutoCAD latest version
- Specifications shall be in Specs Intact Unified Facilities Guide Specifications (UFGS) format or an approved Construction Specifications Institute (CSI) format.

7.0 COMMISSIONING ACTIVITIES

The contractor shall develop and implement a quality-oriented Commissioning Plan for achieving, verifying, and documenting that the performance of the lighting systems, and assemblies meets the defined objectives. The Commissioning process must achieve at a minimum the following objectives:

- Verify and document that equipment is installed per manufacturer's recommendations and to industry accepted minimum standards.
- Verify and document that equipment and systems receive complete operational checkout by installing contractor.
- Ensure lighting levels are per design for emergency lights during power failure
- Ensure lighting levels are per design for normal lights/operation

- Verify and document equipment and system performance.
- Verify completeness of operations and maintenance manuals.
- Ensure that the owner's operating personnel are adequately trained on the operation and maintenance of building equipment.

8.0 PROJECT SCHEDULE

<u>Task</u>	<u>Time Frame</u>
Kickoff Meeting	14 days after Award
Request list of NASA drawings and specifications required	14 days after Kickoff Meeting
Submit Preliminary Scope of Work for NASA approval	21 days after drwgs & specs are provided
Design Development	45 days after NASA approved SOW
Submit Design	30 days after IGA Development
Construction Kickoff	14 Days after Design Approval
POP	90 Days
Close out	30 Days

8.1 Kickoff Meeting

The Contractor shall attend a Kickoff Meeting and Site Walkover at SSC within 14 days of contract award. All key personnel for the design team shall attend. The meeting will be used for introductions, a statement of work review, discussion to clarify uncertainty or identify possible differing site conditions, and development of an action plan to enable completion of tasks to meet the initial design milestone. The walkover will be used to familiarize the Contractor personnel with SSC and the facility to be modified. The Contractor's Project Lead shall e-mail a summary of the discussion and action items status to the NASA Contracting Officer no later than 72 hours following the Kickoff Meeting.

8.2 Bi-Weekly Status Meeting

The contractor shall participate in a bi-weekly conference call with NASA to review the IGA status and address all action items. The conference call will not occur in weeks containing meetings

9.0 SAFETY AND ENVIRONMENTAL REQUIREMENTS

Submit a safety plan in accordance with NASA standards. Safety Plan shall be approved prior to the start of construction. Provide monthly Training and Mishap Exposure Repots. See details: <http://constructionsafety.ssc.nasa.gov/>

Complete a Preliminary Environmental Survey (PES) Form during the design phase. The NASA environmental office will review the PES and provide a response with applicable environmental requirements.

10.0 ATTACHMENTS

- I. NASA Building 4400 Energy Improvement Project PFA